(and, let me emphasize, *it does not*), then indeed we would say that the average position and momentum satisfy the classical equations. But in reality the classical equations are only approximations, good whenever we can replace the average of *dV/dx* by the function of the average of *x*. When is it reasonable to do this? The answer is whenever the *V* (*x*) varies slowly compared to the size of the wave packet. If *V* varies rapidly across the wave packet, the classical approximation will break down. In fact, in that situation a nice, narrow wave packet will get broken up into a badly scattered wave that has no resemblance to the original wave packet. The probability function will also get scattered. Then you'll have no choice but to solve the Schrödinger equation.

Let's look at this point more closely. Mathematically, we've made no assumptions about the shapes of our wave packets. But we have tacitly thought of them as being nicely shaped functions with a single maximum, smoothly trailing off to zero in the positive and negative directions. This condition, though not explicit in our mathematical assumptions, does have a real impact on whether a particle behaves the way classical mechanics would lead us to expect.

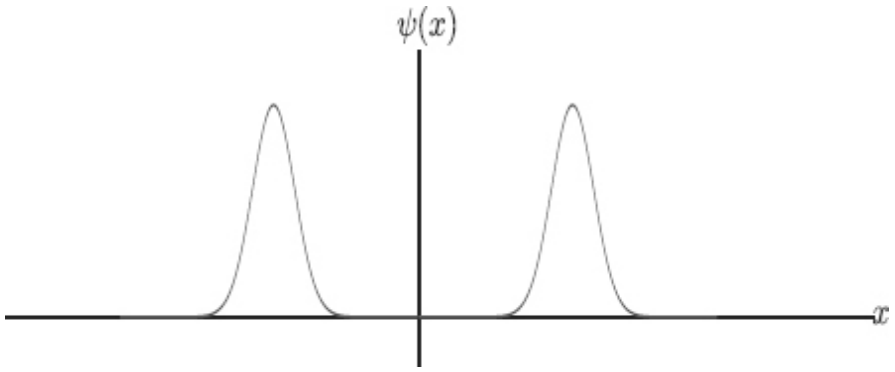


Figure 9.3: Bimodal (Two-Humped) Function, Centered at $x = 0$. Note that $\langle x \rangle = 0$, but $\Delta x > 0$.

To illustrate this point, let's consider a slightly "weird" wave packet. Fig. 9.3 shows a bimodal wave packet (having two maxima), centered at the origin of the *x* axis. Now, let's consider some function of *x*,say *F*(*x*), where *F* represents force. The expectation value of *F*(*x*) is not the same as the function *F* of the expectation value of *x*. In other words,